
BUSINESS REQUIREMENTS

for

LKFORUM WEBSITE

Version 1.0

Prepared by : Quach Gia Kiet
Vo An Khoi

Submitted to : Nguyen Trinh Dong
Lecturer

May 17, 2026

Contents

1	Objective and Scope	3
1.1	Project Objective	3
1.2	Project Scope	3
2	Business Requirements	4
2.1	Application Overview	4
2.2	Domain Model	4
2.2.1	Domain Diagram (Conceptual)	4
2.2.2	Domain Objects Description	5
2.3	Workflow	5
2.3.1	Content Lifecycle & Moderation Workflow	5
2.3.2	Community Management Workflow	5
2.4	Use Cases and Actors	6
2.4.1	Use Case Diagram	6
2.4.2	Description of Actors	7
2.4.3	Description of Use Cases	7
2.5	Security Matrix	9
2.6	User Story	10
2.6.1	Guest Users	10
2.6.2	Members	10
2.6.3	Community Owners & Moderators	11
2.6.4	System Administrators	12
3	Appendix	13
3.1	Glossary	13
3.2	Open Issues	13

1 Objective and Scope

1.1 Project Objective

The primary objective of the LKForum project is to design and build a modern, scalable, and highly interactive community platform. Unlike general social networks, LKForum focuses on autonomous community management, allowing users to create, configure, and moderate their own niche sub-forums (Communities).

The development focuses on delivering a high-quality system that ensures:

- High performance for real-time interactions, including Instant Messaging and Notifications.
- Superior scalability to handle a large and growing number of concurrent communities and users.
- Robust moderation capabilities, integrating both automated (AI-assisted) and manual review processes.
- High standards for security and data privacy to protect user information and platform integrity.

1.2 Project Scope

The project scope includes the following modules:

- **Identity & Access Management:** Multi-factor authentication, SSO, and granular role-based access control.
- **Community Management:** Creation, configuration (public/private/18+), rule-setting, and membership handling.
- **Content Management:** Multi-format posts (text, image, video, poll), nested comments, and draft management.
- **Engagement System:** Real-time voting (upvote/downvote), saved posts, and activity history.
- **Communication Suite:** Real-time 1-1 messaging via WebSockets and an instant notification system.
- **Moderation & Governance:** Report handling, user banning/muting, and content approval workflows.
- **Administration Dashboard:** Global system oversight, analytics, and platform-wide moderation.

2 Business Requirements

2.1 Application Overview

LKForum is a community-driven social network structured around “Communities” (sub-forums).

- **Users** can discover and join communities based on interests.
- **Content** is categorized by communities, ensuring relevant discussions.
- **Decentralized Moderation:** Each community has its own set of rules and a dedicated moderation team (Owners/Moderators), reducing the burden on global administrators.
- **Real-time Experience:** The platform provides instant feedback, live messaging, and notifications.

2.2 Domain Model

2.2.1 Domain Diagram (Conceptual)

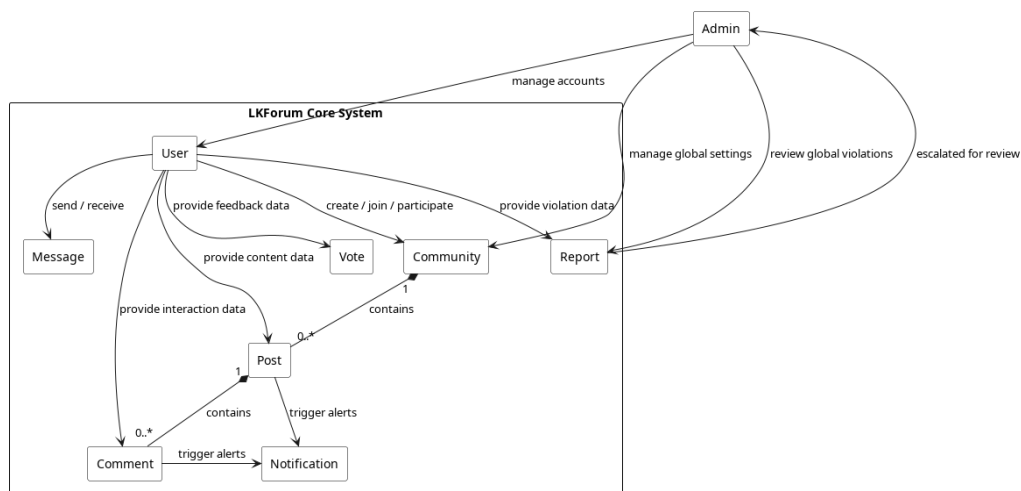


Figure 2.1: Domain Model Diagram

2.2.2 Domain Objects Description

#	Object Name	Object Description
1	User	A registered identity with a profile, karma points, and activity history.
2	Community	A niche hub with specific topics, rules, and privacy settings.
3	Post	Content shared within a community (Text, Image, Video, or Poll).
4	Comment	A response to a post, supporting nested/threaded replies.
5	Vote	An expression of approval (Upvote) or disapproval (Downvote) on content.
6	Message	Direct communication content between users in a 1-1 chat.
7	Notification	Real-time alerts for replies, mentions, votes, or system messages.
8	Report	A user-generated complaint about content or behavior violating rules.

Table 2.1: Domain Objects Description

2.3 Workflow

2.3.1 Content Lifecycle & Moderation Workflow

1. **Creation:** A user creates a post in a community.
2. **AI Screening:** The system automatically checks content for prohibited keywords/media.
3. **Approval (Optional):** If the community requires post approval, the post is held in a *Pending* state.
4. **Moderation:** A Moderator/Owner reviews the post (Approve or Reject).
5. **Publication:** Once approved, the post becomes public and triggers notifications to members.

2.3.2 Community Management Workflow

1. **Setup:** A user creates a community and becomes the Owner.
2. **Configuration:** The Owner sets rules, privacy settings, and invites Moderators.
3. **Governance:** Owners and Moderators monitor reports and can ban/mute users or remove content.

2.4 Use Cases and Actors

2.4.1 Use Case Diagram

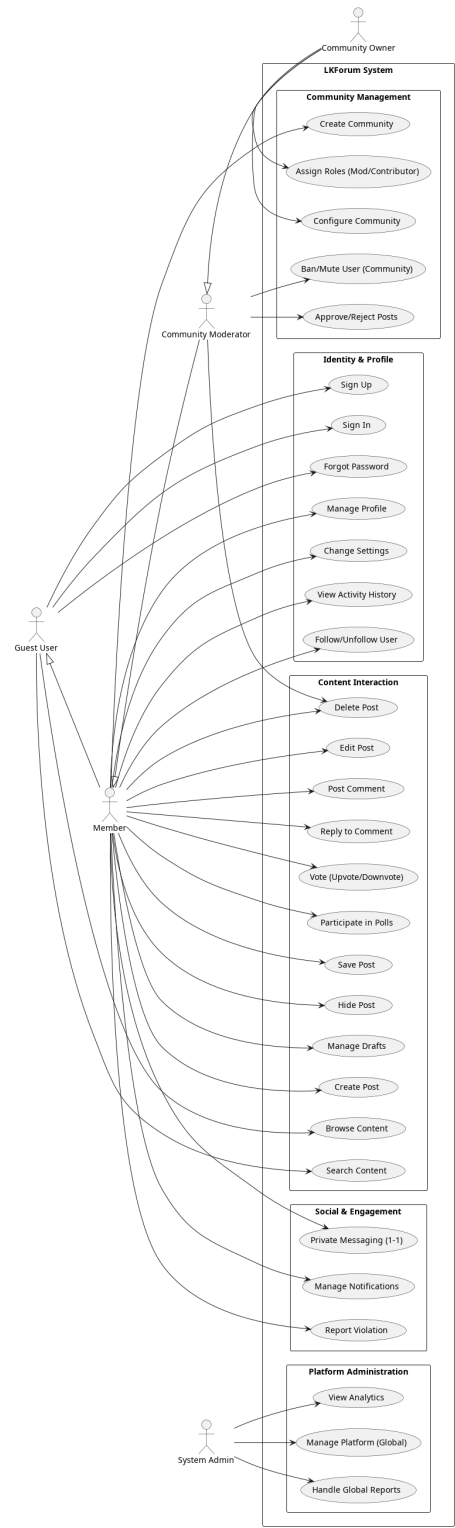


Figure 2.2: Use Case Diagram

2.4.2 Description of Actors

ID	Actor Name	Definition
U-1	Guest User	Unauthenticated user; can browse public content and search.
U-2	Member	Authenticated user; can post, comment, vote, and message.
U-3	Community Moderator	Member with management rights in a specific community.
U-4	Community Owner	Creator of a community; can appoint moderators and change settings.
U-5	System Admin	Global administrator responsible for platform-wide health and safety.

Table 2.2: Description of Actors

2.4.3 Description of Use Cases

ID	Use Case Name	Definition
UC-01	Sign Up	Users create a new account by providing email, username, and password, verified via OTP.
UC-02	Sign In	Authenticated access using credentials or linked Google OAuth account.
UC-03	Forgot Password	Users reset their password through email verification (OTP).
UC-04	Manage Profile	Users update personal information, including bio, avatar, and cover image.
UC-05	Change Settings	Users configure account preferences (privacy, notifications, theme).
UC-06	View Activity History	Users review their past posts, comments, and interactions.
UC-07	Follow/Unfollow User	Users subscribe to another user's activity feed.
UC-08	Browse Content	Users explore communities and view public posts without necessarily logging in.
UC-09	Search Content	Users search for specific keywords, posts, users, or communities.
UC-10	Create Post	Members publish new content (Text, Image, Video) within a selected community.

Table 2.3: Description of Use Cases

UC-11	Edit Post	Members modify the content of their existing posts (stores edit history).
UC-12	Delete Post	Members or Moderators remove posts from public view (soft delete).
UC-13	Post Comment	Users add a response directly to a post.
UC-14	Reply to Comment	Users respond to an existing comment (supporting nested threads).
UC-15	Vote on Content	Users express approval (Upvote) or disapproval (Downvote) on posts/comments.
UC-16	Participate in Poll	Users cast votes in community-created polls.
UC-17	Save Post	Users bookmark posts for quick access later.
UC-18	Hide Post	Users hide specific posts from their personal feed.
UC-19	Manage Drafts	Users save unfinished posts as drafts and resume editing later.
UC-20	Private Messaging	Users send and receive instant 1-1 messages via WebSocket.
UC-21	Manage Notifications	Users view and clear real-time alerts for engagement and system updates.
UC-22	Report Violation	Users flag inappropriate content or behavior for moderator review.
UC-23	Create Community	Users establish a new niche forum and become the “Owner.”
UC-24	Approve/Reject Post	Moderators review pending posts before they appear in the community.
UC-25	Ban/Mute User	Moderators restrict a user’s ability to participate within a community.
UC-26	Configure Community	Owners adjust rules, privacy settings, and appearance of their community.
UC-27	Assign Roles	Owners appoint other members as Moderators or Contributors.
UC-28	Handle Global Reports	System Admins review and act on platform-wide violation reports.
UC-29	Manage Platform	System Admins ban/unban users or communities globally.
UC-30	View Analytics	System Admins monitor platform statistics (DAU, growth, content volume).

Table 2.3: Description of Use Cases

2.5 Security Matrix

#	Function	Guest	Member	Mod	Owner	Admin
1	Sign Up / Sign In / Forgot Password	X	X	X	X	X
2	Manage Profile & Settings		X	X	X	X
3	Follow / Unfollow User		X	X	X	X
4	Browse Public Content & Search	X	X	X	X	X
5	View Own Activity History		X	X	X	X
6	Create Post / Comment / Reply		X	X	X	X
7	Edit / Delete Own Content		X	X	X	X
8	Vote / Participate in Polls		X	X	X	X
9	Save / Hide Post / Manage Drafts		X	X	X	X
10	Send/Receive Private Messages		X	X	X	X
11	Receive Notifications		X	X	X	X
12	Report Violations		X	X	X	X
13	Create New Community		X	X	X	X
14	Approve / Reject / Pin Posts			X	X	X
15	Delete Any Post (In Community)			X	X	X
16	Ban / Mute User (In Community)			X	X	X
17	Configure Community Settings				X	X
18	Assign Roles (Mod/Contributor)				X	X
19	Handle Global Reports					X
20	Global User / Community Ban					X
21	View System Analytics					X

Table 2.4: Security Matrix

2.6 User Story

2.6.1 Guest Users

ID	User Story
US-01	As a Guest User, I want to browse posts so that I can explore content without creating an account.
US-02	As a Guest User, I want to view posts and comments so that I can read discussions.
US-03	As a Guest User, I want to search for keywords so that I can find relevant content.
US-04	As a Guest User, I want to search for communities so that I can discover topics of interest.
US-05	As a Guest User, I want to view user profiles so that I can learn about content creators.
US-06	As a Guest User, I want to be prompted to sign up when attempting restricted actions so that I understand the benefits of creating an account.
US-07	As a Guest User, I want to sign up with my email and verify via OTP so that I can securely create an account.

Table 2.5: Guest User Stories

2.6.2 Members

ID	User Story
US-08	As a Member, I want to sign in using my Google account so that I can quickly access the platform without creating a new password.
US-09	As a Member, I want to sign in using my email and password so that I can access my account securely.
US-10	As a Member, I want to customize my profile with a bio so that I can describe myself.
US-11	As a Member, I want to upload an avatar so that other users can recognize me.
US-12	As a Member, I want to upload a profile banner so that I can personalize my profile page.
US-13	As a Member, I want to create a text post so that I can share content with others.
US-14	As a Member, I want to upload images to my post so that I can share visual content.
US-15	As a Member, I want to create a poll in my post so that I can gather opinions from others.

Table 2.6: Member User Stories

US-16	As a Member, I want to save my post as a draft so that I can edit and publish it later.
US-17	As a Member, I want to upvote posts so that valuable content is highlighted.
US-18	As a Member, I want to downvote posts so that low-quality content is less visible.
US-19	As a Member, I want to upvote comments so that helpful responses are recognized.
US-20	As a Member, I want to downvote comments so that unhelpful responses are less visible.
US-21	As a Member, I want to create, modify, and delete comments under a post.
US-22	As a Member, I want to reply to a specific comment so that conversations remain structured.
US-23	As a Member, I want to receive notifications when someone replies to my post or comment so that I stay updated.
US-24	As a Member, I want to receive notifications when someone mentions me so that I can respond quickly.
US-25	As a Member, I want to follow other users so that I can see their activities in my feed.
US-26	As a Member, I want to view posts from users I follow so that I can stay updated with their content.
US-27	As a Member, I want to send private messages to other users so that I can communicate one-on-one.
US-28	As a Member, I want to report inappropriate content so that I can help maintain a safe community.

Table 2.6: Member User Stories

2.6.3 Community Owners & Moderators

ID	User Story
US-29	As a Community Owner, I want to create a community so that I can build a space for specific topics.
US-30	As a Community Owner, I want to define community rules so that members understand acceptable behavior.
US-31	As a Community Owner, I want to set privacy settings (public or private) so that I can control access to the community.
US-32	As a Community Owner, I want to assign moderator roles to members so that they can help manage the community.
US-33	As a Community Moderator, I want to review and approve pending posts so that only appropriate content is published.

Table 2.7: Community Owner and Moderator User Stories

US-34	As a Community Moderator, I want to remove posts or comments so that I can enforce community rules.
US-35	As a Community Moderator, I want to mute users so that they temporarily cannot interact.
US-36	As a Community Moderator, I want to ban users so that I can prevent disruptive behavior.

Table 2.7: Community Owner and Moderator User Stories

2.6.4 System Administrators

ID	User Story
US-37	As a System Admin, I want to view a dashboard with system analytics so that I can monitor platform performance.
US-38	As a System Admin, I want to review reports escalated from communities so that I can take appropriate action.
US-39	As a System Admin, I want to ban users globally so that malicious users are removed from the platform.
US-40	As a System Admin, I want to remove or restrict communities so that harmful communities are controlled.

Table 2.8: System Administrator User Stories

3 Appendix

3.1 Glossary

- **Karma:** A reputation score based on the net value of upvotes and downvotes received.
- **Sub-forum:** Synonymous with “Community” in LKForum.
- **SSO:** Single Sign-On (Google Authentication).
- **OTP:** One-Time Password sent via email for verification.

3.2 Open Issues

- Implementation of cross-posting (sharing posts across multiple communities).
- Automated AI moderation for image/video content (currently focusing on text).
- Integration of a global search engine (Elasticsearch).